

Table 1: Dataset-specific hyperparameters selected for RetinexTapetum.

Hyperparameter	LOL-v1	LOL-v2 Real	LOL-v2 Synthetic	UHD-LL (down4)
Learning rate	6.9063×10^{-4}	6.6209×10^{-4}	6.9916×10^{-4}	4.6948×10^{-4}
Minimum learning rate	1.4628×10^{-5}	4.0830×10^{-5}	5.0238×10^{-5}	2.1283×10^{-5}
Gradient clipping norm	3	3	8	3
$\lambda_{\max}$	1.80	2.00	1.30	1.35
w_{SSIM}	0.50	0.45	0.65	0.35
w_{LPIPS}	0.10	0.12	0.08	0.08
w_{color}	0.04	0.02	0.14	0.06
w_{chroma}	0.04	0.14	0.12	0.12
w_{edge}	0.07	0.02	0.00	0.04
$w_{\text{dark-noise}}$	0.03	0.01	0.05	0.01
w_{attn}	0.009	0.010	0.007	0.012

Common settings were 120 epochs, batch size 2, crop size 256×256 , 128 base channels, $w_{\text{L1}} = 1.0$, $w_{\text{recon-low}} = 1.0$, $w_{\text{recon-high}} = 0.85$, $w_{\text{reflect}} = 0.06$, $w_{\text{smooth-low}} = 0.08$, $w_{\text{smooth-high}} = 0.08$, and $w_{\text{smooth-enh}} = 0.06$. HPO LPIPS was evaluated after resizing to 512 pixels.

Table 2: Three-seed HPO validation performance of the selected configurations.

Dataset	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
LOL-v1	21.907 ± 0.077	0.8652 ± 0.0022	0.1256 ± 0.0055
LOL-v2 Real	21.576 ± 0.037	0.8317 ± 0.0004	0.1615 ± 0.0059
LOL-v2 Synthetic	25.173 ± 0.045	0.9357 ± 0.0014	0.0554 ± 0.0011
UHD-LL (down4)	27.035 ± 0.178	0.9263 ± 0.0021	0.0670 ± 0.0027

Values are mean $\pm$ standard deviation over seeds $\{42, 123, 3407\}$. Selection used a 0.15-dB PSNR tolerance and a 0.002 SSIM tolerance, followed by minimum mean LPIPS.

Table 3: Training diagnostics and independent test-set performance of the selected RetinexTapetum checkpoints.

Dataset	Seed	Epochs	Loss at best epoch		HPO validation			Independent test split			
			Train	Val.	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	N	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
LOL-v1	42	88/120	0.2779	0.2713	22.009	0.8681	0.1185	15	21.039	0.8316	0.1357
LOL-v2 Real	123	74/120	0.2819	0.2864	21.598	0.8318	0.1532	100	20.019	0.8405	0.1258
LOL-v2 Synthetic	3407	111/120	0.1780	0.1809	25.117	0.9375	0.0538	100	25.053	0.9322	0.0577
UHD-LL (down4)	3407	96/120	0.1587	0.1554	27.259	0.9281	0.0652	150	24.711	0.9115	0.0884

Epochs are reported as best/completed. The training pipeline did not log train-split PSNR, SSIM, or LPIPS; therefore, the only available train-split metric is the loss at the selected epoch. HPO validation LPIPS used 512-pixel resizing, whereas test LPIPS was taken from the paper-metric output; direct validation–test LPIPS differences should therefore be interpreted cautiously.